



Verified Carbon Standard

VERIFICATION REPORT HYDROELECTRIC PROJECT EL EDÉN



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Summary:

ALLCOT AG has commissioned “AENOR CONFÍA S.A.U.” to carry out the Verification of the project - “Hydroelectric Project El Edén”, with regard to the relevant requirements of VCS Standard Version 4.7 /45/.

The “Hydroelectric Project El Edén” is a hydroelectric project plant in the river basin “La Miel” in the east of the Department of Caldas, Colombia, initially implemented by Central Hidroeléctrica El Edén S.A.S E.S.P. It generates electricity by the river “La Miel” through the facilities of a concrete weir and intake, a desander, channel, forebay/settling tank, high-pressure penstock, powerhouse and discharge channel. It has two Pelton turbines with 10.319 MW and two synchronous generators with 11,154 kVA each, connected to a transformer that increases the voltage from 13.2 kV to 33 kV at the project site. The installed capacity of the hydroelectric plant at the interconnection point documented is 19.9 MW /7//10/. The project activity will lead to a reduction of greenhouse gas emissions due to the displacement of fossil fuel-based electricity by zero-emission hydroelectric renewable energy-based electricity.

Proeléctrica S.A.S. E.S.P. and Central Hiroeléctrica El Edén S.A.S. E.S.P. were officially merged on under the supervision of the Superintendency of Public Services, describing an enforceable and irrevocable agreement with the holder of the statutory, property or contractual right over the facilities of the plant and its equipment /6/. In this regard, Proeléctrica S.A.S. E.S.P. is the project proponent (PP) of the project, as informed to Verra on 01/09/2023 through the deed of partial release and the deed of accession, in line with VCS requirements /64-66/. The change has been approved and recognized on Verra Registry.

The purpose of the verification was the independent evaluation of the project’s compliance with the VCS Standard v4.7, the CDM:ACM0002 methodology applied (Consolidated baseline methodology for grid-connected electricity generation from renewable sources, version 13.0) and the requirements of VCS, also the verification applied a strategic and risk analysis required in the norm ISO 14064-3: 2019. The process was performed through a combination of desk review, interviews, on-site visit, and communications with relevant personnel. During the verification 2 CLs and 3 CARs were reported. All these issues where appropriately closed by means of corrections, more clear explanations, and other supporting documents.

AENOR carried out a final verification report and deems the GHG emission reductions have been calculated without material misstatements, in a conservative and appropriate manner, with reasonable level of assurance and that the project complies with all of the verification criteria for VCS. The assessment team has no restrictions or uncertainties with respect to the compliance of the project with the verification criteria.

AENOR concludes that the net GHG emission reductions at verification stage have been quantified in accordance with VCS rules. AENOR assessed the calculations and can confirm that the estimated GHG

emission reductions are correct. The project has calculated the reduction of 81,476 tCO₂e (GHG benefit) for the monitoring period (01/01/2021 to 31/07/2024), through 265,227MWh delivered to the National Interconnected System (SIN).

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1 INTRODUCTION

1.1 Objective

The purpose of the verification audit activity was to conduct an independent assessment of the project in order to determine whether the project complies with the verification criteria, as set out in the guidance documents listed in Section 1.2 of this report.

1.2 Scope and Criteria

AENOR CONFÍA, S.A.U. (Unipersonal) verification/validation entity, is accredited by ANAB with accreditation number AEN 8993. The scope of the verification audit is to verify the reductions of the proposed project activity in Colombia against the Verified Carbon Standard, the identified methodology and associated tools, for the monitoring period from 01/01/2021 to 31/07/2024.

The scope of the verification is the independent and objective review of the Monitoring Report /1/ against the relevant VCS criteria and rules described below, including the monitoring data and other relevant documents available for verification.

The scope was defined as follows:

- The project implementation and operation, and its baseline scenarios as per the registered VCS PD, as well as the physical infrastructure, activities, technologies and processes of the project.
- The data is recorded and stored as per the monitoring methodology.
- The types of GHGs that are applicable to the project and the GHG sources, sinks and/or reservoirs are applicable to the project, without any double counting.
- The project monitoring period and the practiced monitoring system plan, where data reported is accurate, complete, consistent, transparent and free of material error or omission.

The assessment team has employed a risk-based approach to assess the completeness and accuracy of the claims and conservativeness of the assumptions in the VCS MR. The main focus of the assessment team is to identify the significant risks for the project implementation, operation and the generation of VERs/VCUs. The verification is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the monitoring report. The only purpose of the verification is its usage during the registration/issuance process as part of the VCS project cycle. Therefore, AENOR CONFIA cannot be held liable by any party for decisions made or not made based on the verification opinion, which will go beyond that purpose.

The verification assessment was performed in accordance the requirements detailed in Section 4 of the VCS Standard, including the following documents:

- VCS Standard v4.7

- VCS Program Guide v4.4
- VCS Monitoring Report Template v4.4
- VCS Verification Report Template v4.4
- VCS Program Definitions, v4.5
- Validation and Verification Manual, v3.2
- CDM:ACM0002: Consolidated baseline methodology for grid-connected electricity generation from renewable sources, Version 13.0 /46/.
- Background investigation, on-site inspection and follow up interviews.

Furthermore, the auditing team assessed additional documentation by third parties like host party legislation, technical reports, the project design and technical data available on public domain. A desk review has been carried out to assess the following (documentation reflected in the Appendix II of this report):

- Compliance with relevant law and regulations
- Stakeholder comments and signed Environmental Compliance Reports (ECRs)
- System of Requests, Complaints, Claims and Suggestions procedure and records (RCCS)
- Environmental Impact Assessment (EIA), Environmental Management Plan (EMP) and resolutions
- Grid connection contract, Commercial operation declaration and Commissioning Certificate
- Incorporations and withdrawals from employment: 2021 – 2024
- O&M reports
- Training records
- Internal policies and protocols
- Technical specifications of turbines, meters, etc.
- Energy data: 2021 - 2024
- Calibration Certificates.

1.3 Level of Assurance

The assessment was conducted to provide a reasonable level of assurance of conformance against the defined audit criteria and materiality thresholds within the audit scope.

Based on the verification assessment described in the previous section 1.2 and the audit findings, a positive evaluation statement reasonably assures that the project GHG assertions are materially correct and is a fair representation of the GHG data and information.

AENOR applies the rule-based approach aimed at focusing on the fulfillment of the rules determined by the VCS Standard, in order to assure not omitting any part of the mandatory processes. A few discrepancies were found during the verification and the findings were submitted to the project

proponent, indicated under the titles corrective action requests (CARs) and clarification requests (CLs). CARs and CLs require the PP to take relevant actions. Details on the verification work carried out and the verification opinion are presented in the Verification statement in section 5 below this report.

All versions of the verification report were subjected to an independent internal technical review before being submitted to the client to confirm that all verification activities had been completed according to the pertinent instructions required. The technical review was performed by a technical reviewer qualified in accordance with AENOR´s qualification scheme for VCS validation and verification.

Name	Role in the team
Asís Arranz	Lead Auditor
Francy Ramírez	Auditor
Sergio Rodrigo	Auditor in trainee
Luis Javier Arribas	Technical Reviewer

1.4 Summary Description of the Project

The “Hydroelectric Project El Edén” is a project which verts electricity to the National Interconnected System (SIN, for its initials in Spanish) since the 16-February-2017 /7/, through the operation of a hydroelectric power plant in the river basin “La Miel” in the east of the Department of Caldas, Colombia. The project has two Pelton turbines with 10.319 MW and two synchronous generators with 11,154 kVA each, reaching an installed capacity at the interconnection point by 19.9 MW /7//10/. The project activity will lead to a reduction of greenhouse gas emissions due to the displacement of fossil fuel-based electricity by zero-emission hydroelectric based electricity.

It is noticeable that the installed capacity of the plant was based on the design, and it has varied slightly under real conditions to 20.078 MW. These modifications are not considered significant, as they do not affect either the baseline or additionality (Section 3.2 of this report). For the latter case, a sensitivity analysis was included in the registered PD, demonstrating that a variation of +10% of electricity variation would not significantly affect the IRR of the project by 13.32%, being below the 14.90% of IRR benchmark /1-2/.

The project activity is categorized under scope 1 - Energy (renewable/non-renewable) and it is a single project which is not included in a grouped project. The project proponent is Proeléctrica S.A.S E.S.P.

The first crediting period of the project reflected in the PD is for 10 years, from 01/09/2015 to 31/08/2025 /2/. However, as the project activity started operations on a later stage (project starting date 16/02/2017), the actual project activity crediting period is from 16-February-2017 to 15-February-2027, as reflected on Verra’s website. During this monitoring period, from 01/01/2021 to 31/07/2024, the project has generated 265,227 MWh to the SIN, replacing anthropogenic emissions of greenhouse gases (GHG’s) by a total of 81,476 tCO₂ /1//3-5/.

As described in section 1.1 of the MR:

- The Contract for the supply, assembly, and commissioning of electromechanical equipment was signed on 28-June-2013 /8-9/.
- The request for registration in the Public Service Providers Registry (RUPS – for its translation in Spanish) to the Superintendence of Public Domiciliary Services was on 4-November-2016, and approved on 22-November-2016.
- The Certificate of compliance with the Connection Code (Resolution CREG 025 of 1995) was given on 28-February-2017.
- The project start date was on 16-February-2017, when the project activity started generating electricity /7/.

During the on-site inspection of January 2025, location, equipment, technical aspects of the project activity and monitoring arrangements mentioned in the VCS PD and the MR were verified by the audit team, and a deviation was found during this monitoring period (see section 3.2 below). The same was also crosschecked during the desk review of supporting documents like technical specifications, single line diagram and commissioning certificates /9/. The details of project capacity and location details for all the project instances are as follows.

Location	Installed capacity (MW)
Project capacity recognized by the connection contract No. 046 from 2014	19.9
Project capacity by installed generators	20.078

2 VERIFICATION PROCESS

2.1 Method and Criteria

The verification was performed by a combination of desk review, interviews, on-site visit, and communications with relevant personnel, as discussed in Sections 2.2 - 2.4 of this report. At all times, the project was assessed for conformance with VCS requirements and the described in Section 1.2 of this report. As discussed in Section 2.5, findings were issued to ensure the project was in full conformance to all requirements.

The verification activities in which risks were assessed were the evaluations of the applicability, baseline scenario, additionality, leakage, monitoring system, safeguards, etc.

The verification process was conducted using standard auditing techniques to assess the correctness of the information provided by the project participants. Before the assessment begins, members of the team

covering the technical scope(s), sectoral scope(s), and relevant host country experience for evaluating the VCS project activity are appointed.

In order to ensure transparency, assumptions must be clear and stated explicitly and background material must also be referenced. AENOR CONFIA has developed a desk review and the assessment of all the supporting documentation provided by the PP, as well as the references reflected in the MR, in a transparent manner, verifying all the project criteria requirements.

AENOR carried out a deep and meticulous review of the spreadsheets to verify the correct application of the methodology (formulae, equations, etc.) and checked that data required calculating the GHG reductions were appropriately provided /3/. Based on the assessment carried out, AENOR confirms with a reasonable level of assurance that the claimed emission reductions are free from material errors, omissions, or misstatements.

AENOR confirms that sufficient evidence was presented for the ex-post calculated net anthropogenic GHG emission reductions and that there is a clear audit trail that contains the evidence and records that verify the stated figure in this verification report since:

- Sufficient evidence available: The project participant has provided the 100% of data used in the calculations to achieve the final estimated amount of GHG emission reductions.
- Nature of evidence: The raw data was collected from reliable sources. They are detailed in the project documents, were provided to the verification team and checked during the interviews of the on-site visit in January 2025.
- Cross-checked evidence: AENOR cross-checked the collected information through interviews with stakeholders and reproducing calculations and statements.

Hence, AENOR confirms that the stated figures in the MR /1/ are correct and confirms that is able to certify the ex-post net anthropogenic GHG reductions based on verifiable and reliable evidence.

Regarding the verification schedule, the following key milestones are highlighted:

1. Initial documentation receipt (MR and ER spreadsheet): 7th of November 2024.
2. Final Audit Plan: 20th of December 2024.
3. Notice of Verification Services (NOVS): 20th of December 2024.
4. Site visit dates: 15th of January 2025.
5. Issuance of the first round of findings: 24th of January 2025.
6. Closure of findings: 19th of March 2025.
7. Technical Review: 28th of March 2025.

According to the sectoral scope, technical area and experience in the sectoral or national business environment, AENOR has composed a project assessment team in accordance with the appointment rules in the internal Quality Management System.

The four qualification levels for team members that are assigned by formal appointment rules are presented in the table of Section 1.3 of this report above.

2.2 Document Review

A detailed review of all project documentation was conducted to ensure consistency with and identify any deviation from the VCS program's requirements. The Project Monitoring Report (MR) submitted by the Project Proponent (PP) /1/ was reviewed against the approved consolidated methodology and against VCS requirements.

Document reviewed included project ownership evidence /6-10/, such as authorizations, agreements and approvals; stakeholder data reflected in environmental compliance reports, visit reports and questionnaires with involved parties /11-15/; methodologies and tools applied by the project /45-53/; calculations and spreadsheets /3-5/; wildlife monitoring plans and sheets /23-25/, internal policies /26-38/, calibration certificates /39-44/, environmental documentation such as EIA, EMP and ECRs /16-22/, evidence and other supporting documentation /54-66/. The supporting documents that were reviewed are all listed in Appendix II of this report. AENOR cross-checked and compared then with the relevant sections of the MR /1/.

To address the corrective actions and clarification requests that arose from the desk review, the PP revised the initial version of the monitoring report (MR) and developed a final version 6. Responses to Corrective Action Requests (CARs) and Clarifications (CLs) can be found in Appendix III.

2.3 Interviews

The AENOR verification team conducted on-site interviews to confirm selected information and resolve issues identified in the document review. Below, the audit team has disclosed the interview process, identified personnel, their roles, who were interviewed and additional relevant information.

The interview and the identifying process are explained as follows. AENOR carried out a completed and accurate interview on 15th January 2025.

AENOR's audit team defined a preliminary questionnaire to complement and support the desk review process as a starting point to interview each one of the operators on-site. This questionnaire considered Verified Carbon Standards (VCS), and the Free, Prior and Informed Consent (FPIC) values based on the information compiled by AENOR's audit team about the project before the on-site visit /62-63/, during the desk-review of the process. A small sample of these questions has been included on the following page for clarification and transparency purposes. Moreover, and as stated below, AENOR's audit team used these questions for gathering purposes. Several new questions were sporadically and systematically included during the interview process, considering the direction in which discussions were directed.

In Table 1 below, it can be found corresponding people related to the project interviewed by AENOR's audit team. The project area site visit happened the 15th of January 2025, where the on-site interviews took place in the offices of the hydroelectric plant El Edén, in the east of the Department of Caldas. All the information was audited with a reasonable level of assurance, and accuracy was acquired.

AENOR's audit team possess the assistance sheets for all the on-site meetings /62/.

Table 1. Interviewees

Name	Title/Organization/Community	Date/Verification mean
Jhon Freddy Ríos Rodríguez	Environmental Manager, Proeléctrica of the plant	Office of Hydroelectric Plant El Edén in Department of Caldas, Colombia – 15 th of January 2025
Emanuel Ayala	TBS Consultant, ALLCOT	
José A. Ortiz	Resident of La Miel village, Pensilvania	
Sandra Johana Moreno Castro	Recreation and sports, Bolivia Pensilvania district	
Fernando Gómez	Master Operator, Proelectrica	
Martha Rocio Rondón	Resident of Soledad Baja village, Pensilvania	

2.4 Site Visits

The project involves the operation of a hydroelectric project in the host country i.e., Colombia, where it displaces electricity from an electricity distribution system that is or would have been supplied by fossil fuel generating units.

The facility consists in a hydroelectric plant in the river basin “La Miel”, in the east of Caldas Department, Colombia, which contents the following facilities: a concrete weir and intake, a desander, channel, forebay/settling tank, high-pressure penstock, powerhouse and discharge channel.

Specific features of the on-site inspection can be seen in Table 2 below.

Table 2. On-site interviews information

Date of site visit: 15-January-2025		
Activity performed on-site	Site location	Team member
- An assessment of the desk review and project documentation on the PD, MR and supporting documentation provided by the PP. - An assessment of the project design and technical specification, project location, implementation status and operation of the project activity as per the VCS PD and MR. - A review of information flows for generating, aggregating and reporting the	Office in the hydroelectric plant El Edén, Department of Caldas, Colombia	Ms. Francy Ramírez

monitoring parameters. • Interviews with relevant personnel to determine the operational and some data collection procedures as per the VCS MR. • An identification of quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters. • A review of calculations and assumptions made in determining the GHG data and emission reductions.		
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2.5 Resolution of Findings

All documentation provided by the PP was assessed against the applicable version of the relevant VCS guidance document. Several clarification requests (CL) and corrective action requests (CAR) were raised and submitted to the PP, which addressed them either by providing to the audit team with the requested information or by making the appropriate corrections. Updated versions of the documentation were submitted by the PP and the audit team reassessed them against the guidance documentation. This process was repeated iteratively until all CL and CAR were fully closed. Specifically, 2 CLs and 3 CARs were raised during the VCS findings phase of the auditing process.

All findings issued by the AENOR audit team during the verification process have been closed. In accordance with VCS Standard v4.7, all findings issued during the verification process and the inputs for their closure are described in Appendix III of this report.

2.5.1 Forward Action Requests

One FAR has been raised during the current verification period due to unavailability of the PP to apply the RENARE registration. This is because the website of RENARE has problems and thus, the PP is requested that, as an integral part of the verification plan for the future verification periods, a specific assessment designed would be required to confirm the absence of double counting or double claiming associated with Colombia VCS projects.

2.6 Eligibility for Validation Activities

The validation/verification body AENOR CONFIA has not undertaken validation activities as part of the verification of the project.

3 VALIDATION FINDINGS

No validation activities have been performed during the current monitoring period.

3.1 Methodology Deviations

The verification team confirms that the registered VCS MR complies with the requirements in the applied monitoring methodology ACM0002 version 13.0.0 /46/. As a result of the documental review, it is concluded that there are no methodology or project description deviations belonging to this monitoring period.

3.2 Project Description Deviations

As indicated on previous section 1.4 of this report, the installed capacity of the plant was based on the design, and it has varied slightly under real conditions from 19.9 MW /7//10/ to 20.078 MW. Although this issue was verified during the previous verification, it has been reviewed during this monitoring period, where the audit team verified the name plates of the installed turbines, and the rated power output turbine is 10,319 kW (i.e. 10.319 MW) per equipment, during the on-site visit of January 2025. These modifications are not considered significant, as they do not affect either the baseline or additionality (Section 3.2.2 of the MR). Additionally, a sensitivity analysis was included in the registered PD regarding this issue, demonstrating that a variation of +10% of electricity variation would not significantly affect the IRR of the project /2/.

Also, as verified in the previous verifications, the initial starting operation date of the project changed from 01/01/2015 to 16/02/2017, and the starting date of the crediting period from 01/09/2015 to 16/02/2017.

On the other hand, the Project Proponent (PP) was modified in 2023, during this monitoring period. The PP signed a Deed of Partial Release dated on 01/09/2023 /66/, where the former PP CENTRAL HIDROELÉCTRICA EL EDÉN S.A.S E.S.P. was formally released from the representations, warranties, obligations, and liabilities assumed under the VCS Registration Deed of Representation /65/, which was originally executed on 20/09/2013. The change has been approved and recognized on Verry Registry as PROELECTRICA S.A.S. E.S.P.

It has been concluded that project description deviations described do not affect the applicability of the methodology, additionality and the appropriateness of the baseline scenario and hence it is valid and accepted.

3.3 New Project Activity Instances in Grouped Projects

The project activity is not a part of a grouped project. Therefore, this section is not applicable.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☐ Yes

☒ No

4 VERIFICATION FINDINGS

4.1 Project Details

Item	Evidence gathering activities, evidence checked, and assessment conclusion:
Audit history	Validation: 23-September-2013 First Verification Period: 16-February-2017 to 31-December-2020 Second Verification Period: 01-January-2021 to 31-July-2024
Double counting and participation under other GHG programs	The VVB has verified the ownership of the PP through the on-site visit and evidence gathered /6-10/, and deems Proelétrica S.A.S. E.S.P. and Central Hidroeléctrica El Edén S.A.S. E.S.P. were merged under the supervision of the Superintendency of Public Services, describing an enforceable and irrevocable agreement with the holder of the statutory, property or contractual right in the plant and its equipment. Therefore, Proelétrica S.A.S. E.S.P. is the project proponent of the project, and it was informed to Verra on 01/09/2023 through the deed of partial release and the deed of accession forms, which have been approved and recognized on Verra Registry /64-66/. Also, the audit team has checked whether the project receive, seek, participate or has been rejected by other GHG program, and confirms it is in line with the double counting requirements of VCS Standard (p.3.23). It is concluded that the project activity is not involved on other Emissions trading programs and other binding limits. The assessment team reviewed other standards and confirmed that the Net GHG emission reductions or reductions generated by the project will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions in any Emission Trading program or other binding limits.
No double claiming with emissions trading programs or binding emission limits	N/A
No double claiming with other forms of environmental credit	N/A

Supply chain (scope 3) emissions double claiming	The project proponent is not a buyer or seller of any product whose emissions footprint is changed by the project activities /8/.
Sustainable development contributions	The project activity has many benefits and contributes to the following sustainable development (section 1.12 of the MR): • <u>SDG 13.2</u>: The project replaces fossil-fuel based power generation with clean energy through water. The estimates for the monitoring period are in total 81,476 tCO₂e emission reductions over three years and seven months, and 321,024 tCO₂e during the 10-years crediting period. The PP monitors this indicator as per the renewable energy generated in MWh /2-3/. • <u>SDG 8.5</u>: The project requires the maintenance of the hydroelectric plant facilities that will require outsourced services, leading to increases in the green economy: 16 jobs during the monitoring period. Employees contracts have been provided to the VVB, and the PP monitors it over the whole crediting period. The audit team verified during the on-site visit the procedures applied by the staff members /55//61/. • <u>SDG 7.2</u>: The project displaces emission intensive grid electricity through the generation of zero-emission hydroelectricity, contributing towards improving access to a reliable, clean source of energy in Colombia. During this monitoring period, the project has generated 265,227 MWh, and estimated 1,045 GWh over the 10-years of the crediting period. The PP monitors the electricity generated in MWh /4-5/.
Additional information relevant to the project	The project activity does not apply leakage management and there has not been any exclusion of commercially sensitive information from the public version of the monitoring report. Finally, there are no further additional relevant legislative, technical, economic, sectoral, social, environmental, geographic, site-specific and/or temporal information that may have a bearing on the eligibility of the project, the net greenhouse gas emission reductions.

Hence, in view of the assessment of VCS MR /1/ and supporting documents listed in Appendix II of this report, the verification team is able to confirm that the description contained is complete and accurate. The VCS MR complied with relevant forms and guidance for completing the VCS MR.

4.2 Safeguards and Stakeholder Engagement

4.2.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder identification	The VVB has reviewed and assessed the MR /1/ of the project and all the supporting documentation of the project, as well as the PD /2/, and confirms the stakeholder identification has not changed since the validation.
Legal or customary tenure/access rights	After reviewing the supporting documentation provided by the PP related on RCCS, Environmental Compliance Reports and Environmental Impact Assessment resolutions /12-14//18/, the VVB confirms the project does not affect the territory of local communities or Indigenous Peoples, and there no collective or conflicting rights related to the area have been identified. Therefore, no tenure, access rights to territories and resources are needed to be considered.
Stakeholder diversity and changes over time	The audit team confirms there have not been changes in stakeholder diversity, on the following /12-14//18/: - Local and national governments, responsible for regulating and overseeing the project. - Local communities located in the project's area of influence. Nonetheless, as described in section 3.2.2, it is noticeable that the project participant of the project has changed to Proeléctrica S.A.S. E.S.P., as responsible for the operation of the hydroelectric plant /64-66/.
Expected changes in well-being	There are no indications in the MR and the supporting documentation provided by the PP on changes in well-being and other stakeholder characteristics expected under the baseline scenario /1//12-14//18/. In the absence of the project, the following ecosystem services in the La Miel River basin would continue being deteriorated due to anthropogenic pressure and natural conditions in the area: - Soil regulation and erosion driven by the area's topography, deforestation, road construction without revegetation, and heavy rainfall, affecting landslides and the ecosystem stability.

	- The physical and chemical conditions of the water would progressively deteriorate due to uncontrolled water extractions and untreated wastewater discharges, impacting both biodiversity and water availability. - Transformation of forests into crops, due to pasture and sugarcane, increasing the demand for wood for agricultural infrastructure. Only forests in inaccessible areas could be partially preserved. In addition, some endemic species could face local extinction. - Degradation of landscape would continue due to population and resource demand growth. If not for adequate roads, some communities would be isolated, being unable to develop socially and economically.
Location of stakeholders	There have been not changes on the location of local stakeholders of the project, being still located in the following communities around the Caldas department in Colombia: Pensilvania, Manzanares and Marquetalia /1/.
Location of resources	There have been no changes of the location of resources /1/.

4.2.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Ongoing consultation	The mechanisms and mapping for ongoing communication with stakeholders have been revised with the Environmental Compliance Reports (ECRs) and records, where stakeholder inputs can be revised /12-14/. These outline the scope and procedures for ongoing stakeholder engagement and mapping regarding the hydroelectric plant operations and activities. Regarding the grievance redress procedure, the audit team assessed its compliance with Section 3.18.4 of the VCS Standard v4.7 /45/. The project has a Grievance Mechanism Procedure (section 2.1.4 of the MR) to receive and facilitate resolution of affected communities' concerns and grievances about the hydroelectric plant environmental and social performance /32/. It includes well defined and open channels for reciprocal communication and build the capacity of stakeholders to be involved in these processes, the effectiveness and

	appropriateness of the engagement processes. Results are monitored, evaluated and included in the ECRs, by analysing stakeholder feedback during engagement activities, using an analysis procedure that will be developed to evaluate aspects of inclusivity, engagement modality, means and methods for disclosure of information and their appropriateness for the target stakeholder group, and stakeholder participation.
Date(s) of stakeholder consultation	The following dates of stakeholder consultation have been already verified in previous verification process with VCS. In addition, the VVB received evidence of the first round of stakeholder consultation in August 2010 /15/: 1. August 2010 to May 2011 2. June to August 2011 3. March to May 2012 4. 14-April-2013
Communication of monitored results	The auditing team has evaluated the procedure applied by the project for the receipt and response of requests, complaints, claims, compliments, and suggestions (RCCS) through the corresponding internal policy of the project, as well as ECRs and records /12-14//32/.
Consultation records	The audit team has evaluated the Environmental Management Plan (EMP) report of the project during the operation phases /19/, as well as management plan sheets provided by the PP for the following items: 1. Biotic: Compensation and restoration of flora, protection and conservation of habitats, and monitoring of endangered and/or endemic species that may be found in the area of direct influence /20/. 2. Physic: Monitoring and tracking of soil resources, disposal and treatment of solid and liquid waste, water and air resources /21/. 3. Social: Community and institutional information and participation, information and training for project-related personnel, support for institutional management capacity, information and awareness-raising about the project for the surrounding community, local labor recruitment program, archaeological prospecting, impact on third parties, and social and property management /22/.

	The VVB confirms the veracity of the execution of the studies described, implementing measures with estimated costs, where community and third parties sensibilization with the hydroelectric project and its different stages, is shown.
Stakeholder input	Compliance Reports presented to Corpocaldas (Corporation of the Autonomous Regional Government of Caldas) by the project have been provided to the VVB /12-13/, evidencing operation, maintenance plans and compliance with the Environmental License. As described in Section 2.1.1 of the MR, personnel responsible for the environmental activities, updated schedules and stages of the environmental and social plans and programs are represented /17/. Comments of public participation, objectives and risks of the project are disclosed in the reports, regarding risks to stakeholder participation, working conditions, safety of women and girls, safety of minority and marginalized groups, including children and training socially excluded people and pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials) /12-14//32/. Overall, there were some comments to the hydroelectric facility and each comment was responded appropriately. The audit team considers that the approach maintained by the PP demonstrate that local population are involved in project activities, and that project stakeholders clearly understand their roles and responsibilities, their voice and importance and their ability to release grievances following a formal procedure as described in the grievance redress mechanism. The overall conclusion of the local stakeholder consultation is that communities perceived the project as having a positive socio-economic and environmental impact. In general, communities perceived project activities as an important driver for development of surrounding communities. Hence, in the opinion of the AENOR's audit team the local stakeholder consultation process was suitability performed and the PP's response to the inputs was appropriate. The audit team deems that the PP communicated the information about the project design and its operations, risks, costs and benefits, relevant laws and questions and the process of VCS Program verification are in accordance with the requirements established by the VCS Standard.

4.2.3 Free, Prior and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Consent	As commented in the previous section, the project has environmental authorization after compliance reports were presented to Corpocaldas /16-18/. There are no property rights' holders who are not project proponents affected by the project. Nonetheless, a consent document was shared from AENOR's audit team during the interviews of the on-site visit, and signed for all the corresponding participants /62-63/. The audit team has verified that the facility is owned by the Proeléctrica S.A.S. E.S.P., so no additional consent regarding access or its use is required. In addition, the project has an established Informed Consultation and Participation process for projects with potentially significant adverse impacts on affected communities /32/.
Outcome of FPIC discussion	The project has the approval of the environmental authorization and has demonstrated through the ECRs that the project does not exacerbate nor influence the outcomes of unresolved conflicts /12-13//17/. The audit team reviewed the arguments provided for the stakeholders during past rounds, and deems they were reasonable /11//14-15/.

4.2.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Grievance received and steps taken to resolve the grievance including the outcomes of the resolution	The PP has provided several lists of Requests, Complaints, Claims and Suggestions (RCCS) records, where all kind of grievances received about the project from agents and stakeholders can be observed, as well as the measures taken to resolve them. The comments reflected in the MR on Section 2.1.4, on the dates 01/06/2021 and 06/08/2021 have been corroborated by the VVB /14/.
Grievance redress procedure	The audit team has assessed the Stakeholder Grievance Mechanism in the receipt and response to requests, complaints, claims, congratulations and suggestions internal procedure report, and deems it includes processes for receiving, hearing, responding and attempting to resolve grievances within a reasonable time period,

taking into account culturally appropriate conflict resolution methods /14//32/.

In the latter report, it is clear the different ways of communication to rise complaints, claims or suggestions, the different phases of the process, and the main responsible persons and their roles to manage the grievance mechanism.

The project has ongoing communication with stakeholders to address disputes that may arise during project implementation and operation:

1. The Environmental and Social Leader will review the requests received at the end of each week to promptly address the requests, complaints, claims, compliments, and suggestions submitted by the various stakeholders.
2. The Social or Socio-Environmental Professional will review the requests received daily to promptly address the requests, complaints, claims, compliments, and suggestions submitted by the various stakeholders. Depending on the type of request received, they will forward it to the appropriate person, ensuring an initial response (confirmation of receipt) to the requester.
3. The Organizational Development and Well-being Leader will review the requests received by email at the end of each week to promptly address the requests, complaints, claims, compliments, and suggestions submitted by the various stakeholders. Depending on the type of request received, they will forward it to the appropriate person, ensuring an initial response (confirmation of receipt) to the requester.

As part of ongoing consultation, the following items are communicated, among others as required:

1. Responses to requests, complaints, claims and suggestions.
2. Issuance of certifications.
3. Requests for the purpose of obtaining a copy of documents.
4. Consultations regarding contract management made by individuals or public entities.
5. Request for information requested by senators, representatives exercising political control, government entities or oversight bodies.

In addition, AENOR audit team reviewed the Verra Registry, particularly the “Other Documents” section, and no public comments have been received by AENOR.

4.2.5 Public Comments

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
Extend to which the non-qualified workers required for the construction will be from the region	All non-qualified workers used for the construction are locally hired.	Non-qualified worker contract locally hired was provided by the PP /12//14//55/.
Improvement of the roads in the area.	As reflected in the management plan report and ECRs, roads are managed for maintenance in the close areas to the project.	Supporting documentation was provided by the PP, as the management plan report and ECRs /12//14/.
Availability of water after implementation of the project and requirement to sign contracts for water security.	The project has environmental license approved by COROCALDAS based on the detailed studies. Besides, the PP has provided an environmental management plan and ECRs that define the required actions to assure minimization of these impacts.	Supporting documentation was provided by the PP, as the management plan report and ECRs /12//14//25/.

The audit team interviewed relevant operators of the plant during the on-site visit of January 2025 (see sections 2.3 and 2.4 of this verification report for more information), where it was reviewed the possible risks and impacts of the project, as well as comments and feedback received from stakeholders and the RCCS reports.

Thus, AENOR's audit team coincides that the project has plans and processes to ensure the project do not create negative impacts on project stakeholders and if negative impacts occur the PP mitigates such impacts where necessary.

4.2.6 Risks to Local Stakeholder and the Environment

4.2.6.1 Management Experience

The management team of the project possesses the requisite expertise and experience in the implementation and operation of hydroelectric plants, renewable energy projects and engaging with

communities. In addition of the latter, the project establishes ongoing communication channels to engage stakeholders and address concerns where these are raised /32/.

The audit team of AENOR was capable to interview during the on-site visit of January 2025, operators of the technical staff that were working in the hydroelectric plant of the project. The auditor was not only able to take notes and pictures of the equipment as the electric meter, but also to verify the software's used in their computers in the substation of the plant, where operators are in constant communication and control through the staff team, for maintenance and security of the area. The operators use phones and radios to maintain contact. The staff archived electronically the corresponding shifts, where the incidents log history is registered, from when they are registered until they are attended and resolved. Also, the technical staff have proper training and Excel files where qualifications are followed up by superiors /54//57-60//62-63/.

4.2.6.2 Risk Assessment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Natural and human-induced risks to stakeholders' wellbeing	The audit team has checked the RCCS records and can confirm comments on risk identified by the project on the availability of water after implementation /14/. On this regard, the program reflects clearly the management of ecological flow conservation as part of the environmental management plan provided, in order to prevent extractions or manipulations that exceed the allowed limit for water capture, and compromise the ecosystem, its native species and services for society /19//21//25/.
Risks to stakeholder participation	The audit team did not identify risks on the stakeholder participation due to the ongoing stakeholder engagement and mapping of the Hydroelectric Project El Edén operations and activities that maintain constant communication channels used to engage stakeholders and address concerns where these are addressed. The VVB has been able to assess the main procedures applied by the PP through supporting documentation and deems that the guiding principles of the Stakeholder Relationship are ethical and transparent /14//32/. Additionally, the Hydroelectric Project El Edén has social programs stated in the Environmental Management Plan (EMP), and Operation and Maintenance Plan (O&MP), both guaranteeing different forms of participation /19//22//54/:

	I. The program “PMA-MS-01. Community and institutional information and participation”, evaluates the understanding of the information provided about the hydroelectric project and its characteristics during the different stages of the project. II. “PMA-MS-06. Impact on third parties”, follows up on the requirements made by the populations within area of influence of the project.
Working conditions	The audit team has checked some comments in the RCCS reports about occupational risks linked to the operation and maintenance tasks /14/. The Hydroelectric Project El Edén has internal policies to mitigate these risks and guarantee good working conditions. The PP has provided plenty supporting documentation such as guidelines for corporate social responsibility, work disconnection policy and environment management of contracts /27-29//31//33-34//37-38/.
Safety of women and girls	The project ensures the safety of women through a strict protocol and control on working behavior. The Hydroelectric Project El Edén has internal policies on human rights, codes of ethics and gender equality, where monitoring and follow-up plans help minimize the risks to the safety of all nearby communities, including women and girls, that could be caused by the operation of the power plant /27-29//31//33-34//36-38/.
Safety of minority and marginalized groups, including children	The project has monitoring and follow-up plans, to engage effectively with vulnerable groups, including children and marginalized groups, that could be caused by the operation of the power plant. PP has provided to the auditing team non-discrimination and child labor prevention policy to ensure prevention of any harm /27-29//31//33-34//35-38/.
Pollutants (air, noise, discharges to water, generation and release of hazardous materials and chemical pesticides and fertilizers)	The audit team confirms that the project has monitoring, and many follow-ups plans on the environment, including the following /16/18-19//25/: - Air resources quality, noise and control of the atmospheric emissions, by determining the concentrations of particulate matter, gases and vapors, in compliance with the established

	Colombian legislation: Resolution 0601 of April 2006 of the MAVDT (Ministry of Environment, Housing and Territorial Development) and Resolution 948 of 1995 Colombia. - Aquatic ecosystem resources, by monitoring the watercourse of tributaries in the section between water intake and discharge from the hydroelectric plant. - Integrated management of hazardous solid waste, and the disposal of liquid waste, fuels, and oil, through the environmental management plan, as well as the through the installation of individual septic tanks and a water treatment plant. - Pesticides and fertilizers are not used in the project.
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4.2.7 Respect for Human Rights and Equity

4.2.7.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination	The audit team confirms that the Hydroelectric Project El Edén has internal policies with serious commitment to a zero-tolerance policy against discrimination and sexual harassment, through a strict protocol of ethical codes. The company commits by ensuring voices through the grievance mechanism for all affected stakeholders and workers /27-29//31//33-34//35-38/.
Sexual harassment	As indicated above, the Hydroelectric Project El Edén has a commitment to a zero-tolerance policy against discrimination and sexual harassment included /27-29//31//33-34//37-38/.
Gender equity in labor and work	As previously indicated on Section 4.2.6.2, the project has implemented gender-sensitive policies to ensure equal opportunities in fair pay in labor. The VVB has verified the supporting documentation provided by the PP and deems it has regular monitoring of gender equity metrics and reporting /27-29//31//33-34//37-38/.
Forced labor	The audit team confirms that the Hydroelectric Project El Edén adheres to international and local labor laws prohibiting forced labor. The non-discrimination, child labor policies and social code of ethics articulates duties to encourage customers, business partners and suppliers of goods

	and services to adopt principles and practices that are comparable to their own /27-29//31//33-35//37-38/.
Child labor	The same way as above, the PP has provided to the audit team internal policies prohibiting the use of employees under the minimum age. /27-29//31//33-35//37-38/.
Human trafficking	Modern slavery is a crime in Colombia and the Hydroelectric Project El Edén takes it seriously, where it is totally prohibited any type of forced labor or human trafficking in its conduct protocols /27-29//31//33-35//37-38/.

4.2.7.2 Human Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risks on human rights identified	The audit team has not identified risks on human rights, being able to verify that the Hydroelectric Project El Edén is firmly committed to the Universal Declaration of Human Rights, ensuring careful adherence within the company. The internal policy shared with the audit team as Code of Ethics, Conduct and other corporate policies, show employees and executives are responsible for acting in accordance with the established guidelines /27//36-37/. In addition, the program PMA-MS-04 of the project, “Information and sensibilization of the project to the local community”, informs the population of surrounding areas, about the activities of the project, during the studies, construction, and operation phases /22/. The documents provided by the PP state that the project is committed to compliance with Human Rights, the International Charter of Human Rights, the conventions of the International Labor Organization (including the Convention 169) and the principles of the Global Compact and as established in the Political Constitution and the international agreements ratified by Colombia (including the United Nations Declaration on the Rights of Indigenous Peoples) /26//30/. The VVB deems the Hydroelectric Project El Edén has constant communication with interested agents to identify the most suitable mitigation strategies on human rights if arise /14//32-33/.

4.2.7.3 Indigenous Peoples and Cultural Heritage

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risks identified on indigenous people or cultural heritage	The audit team has not identified risks on this section, being able to verify that the Hydroelectric Project El Edén is firmly committed to recognize, respecting and championing the rights of Indigenous Peoples (IPs), local communities (LCs), and customary rights holders. The project provides training courses to the personnel linked to the project since the construction phase, where aspects related to the protection and management of archaeological heritage (such as pre-Hispanic artefacts and materials from communities that settled in the area) /58/. The project is committed to, in the event of finding archaeological materials or other elements that suggest belonging to pre-Hispanic communities, the worker or the person who locates the finding must: I. Leave the discovered material the place and inform immediately to the corresponding supervisor/superior. II. The supervisor will report to the field assistant. Apply the Archaeological Survey, Rescue, and Monitoring form. This is verifiable through the Environmental Impact Assessment, Environmental Management Plan and ECRs provided to the VVB /12//16//19//22/. Also, the PP has provided robust framework aligned with international human rights law, mainly the United Nations Declaration on the Rights of Indigenous Peoples and the principles outlined in ILO Convention 169 on Indigenous and Tribal Peoples. This framework underscores our dedication to upholding the principles of free, prior and informed consent (FPIC) /26//30/. The VVB deems the Hydroelectric Project El Edén has constant communication with interested agents to identify the most suitable mitigation strategies and support the

overall well-being of the surrounding communities /32-33/.

4.2.7.4 Property Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risks identified on property rights	The auditing team has verified that the project implements measures to respect and preserve the property rights of individual property owners through the legal frameworks aligned with international human rights standards /6//17-18/. As described in previous sections, the Hydroelectric Project El Edén has a grievance mechanism that addresses disputes or concerns in all the project phases /32/.

4.2.7.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Summary of the benefit sharing plan	The Hydroelectric Project El Edén does not encompass benefit sharing from carbon credits. The audit team concludes that there is no distinct between a benefit-sharing plan associated with carbon credit revenue as there is no impact on property rights, as described in sections 2.1.3 and 2.3.4 of the MR. The VVB has not identified stakeholders with interests directly affected by the project /14/. N/A
Benefit sharing during the monitoring period	N/A

4.2.8 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion
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Impacts on biodiversity and ecosystems	The Hydroelectric Project El Edén identified the following risks on impacts on biodiversity and ecosystems: • Alteration of natural flora populations. • Interruption of biological corridors. • Habitat fragmentation. • Alteration of the ecological function of the forest. The auditing team has verified the project has the following Monitoring and Follow-up Plans (PMS) and Environmental Management Plans (PMA) to mitigate the identified risks: In the PMS /23-25/: 1. Program 1, on monitoring of flora compensation and restoration: assess the progress of protective reforestation established as compensation for the El Edén project. Development of recent associated actions, such as the setup and management of two nurseries at the camp office and the intake area, rescuing the regeneration of forest species. Several of these specimens are donated to various local organizations. In the MR, there are mentioned two recent initiatives: One (a), from the community from the La Soledad village. This group, organized within the village aqueduct, has worked to protect the supplying microcatchment by planting six hundred forest individuals, to mitigate the decreased flow of the river due to the impacts of the El Niño phenomenon. Two (b), the establishment of Bamboo Groves as a compensatory measure. 2. Program 2: habitat protection and conservation, Linear Park PMS_MB_02: Its goal is to assess and maintain the linear park of El Edén. Within the framework of the Project, routine maintenance actions (spraying, mechanical and manual, and spraying) are carried out on the ecological trail in the catchment area. In the PMA /19-21/: Program of Environmental Education: PMA_MB_03_P1. It contributes to the protection, environmental conservation and reduction of negative direct interaction with wildlife and its habitat (hunting, alteration of terrestrial habitats, pollution, etc.). Measures as the following are established and described in the PMA: 1. Conduct scheduled talks in the La Miel River basin, to raise awareness among the population directly involved with the project. In these meetings, workers and other people involved in the construction are informed about restrictions on hunting wildlife,
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	the importance of wildlife in the ecosystem and human activities, the differences between venomous and non-venomous snake species, and logical explanations for beliefs (myths) related to certain animal species and why they are pursued. 2. Distribution of free illustrative pamphlets between students in schools and communities, that guide actions aimed at conserving nature: waste disposal, conservation of riverbank buffers, responsible use of wildlife, and use of flora.
Soil degradation and soil erosion	The Hydroelectric Project El Edén identified the following risks on impacts on soil degradation and soil erosion: • Impact on soil stability and current land use • Alteration of the landscape The project includes in the Environmental Management Plan (PMA) the implementation of the following measures to mitigate the identified risks, which have been revised by the VVB /19//21/: - Control of geotechnical stability and slopes management due to erosion in the different construction fronts of the project. The clearing and stripping are carried out only in the areas strictly necessary for the adaptation of access routes and the construction of both preliminary and final works. - Removal of the organic soil layer in excavation sites conducted by extracting the topsoil along with the vegetation layer at the time of stripping. Materials are stored with full coverage to prevent the effects of water and wind. - Earthworks are carried out to balanced cut and fill volumes, minimizing excavation surpluses. - Maintenance and cleaning of the drainage systems installed on the slopes, to avoid their deterioration and malfunction. - Profiling of the slopes to prevent the activation of erosive processes or mass wasting. - If cracks are found on the surfaces of the slopes, sealing them to prevent the infiltration of surface water or runoff. - Mulching of slopes to help control stability and prevent erosion that could be caused by water action. - Revegetation and reshaping of the intervened areas.

	- Native species are used in the revegetation whenever possible.
Water consumption and stress	The verification team has checked the Hydroelectric Project El Edén identified the following risks on impacts on water consumption and stress: • Impact on water quality due to increased particulate matter and contaminants • Hydrological changes in the section between intake and discharge • Impact on microbiological resources • Alteration of the water resource due to channel diversion, vegetation removal, excavations, domestic installations, and machinery maintenance Where the project includes the following measures in the PMS and PMA to mitigate the identified risks: In the PMS /23-25/: The project monitors the tributary watercourse between the intake and discharge of the hydroelectric plant (PMS_MF_05_P1). On one hand, the project takes actions for the conservation of the La Miel River within the direct influence area of the hydroelectric plant. On the other hand, it prevents alterations in the tributary watercourses and the reduction of their flow contribution to the main channel. In the PMA /19//21/: Through the program 3, the project monitors and control the water resources and its quality, by assessing the physicochemical and hydrobiological conditions of the La Miel River (such as water temperature, dissolved oxygen, conductivity, and pH), at sampling points defined by the watercourses potentially affected by the construction of access roads, main and secondary works, and the operation of camps, workshops, and facilities of the project. The laboratories and the activities carried out follow the IDEAM Guidelines (initials in Spanish), Institute of Hydrology, Meteorology and Environmental Studies, for water monitoring and tracking in all sampling and analysis processes. The program reflects the management of ecological flow conservation as part of the environmental management plan provided, in order to prevent extractions or manipulations that exceed the allowed limit for water capture, and compromise the ecosystem, its native species and services for society.

4.2.8.1 Rare, Threatened, and Endangered species

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Species and habitat	As indicated in section 2.4.1 of the MR on rare, threatened, and endangered species, the EMP of the project establishes in the Program 4 the management of endangered and/or endemic species in the direct influence area, where there have been positive results. In this program, it can be observed the monitoring and collecting information of the populations of the Frog <i>Rheobates palmatus</i>, the Salamander <i>Bolitoglossa ramosi</i>, and the Lizard <i>Anolis tolimensis</i>, endemic species to Colombia, with the goal of prevent and mitigate the project's impact on them. The project proposes the following in summary /19-20//23-25/: - Conduct periodic censuses (during rainy and dry seasons) in the intervention area during the pre-construction phase, to determine population size, habitat use, and the specific requirements of the species. - Assess if the habitat intervention is appropriate or not for the species and determine the management of the area that should be applied based on presence data. Compensation purposes are considered if required. - Periodic monitoring of the species, during construction and operation. For the amphibian population, the species <i>Rhinella humboldti</i> was identified as new. Additionally, the presence of other endemic species was previously found in the monitoring, such as <i>Rheobates palmatus</i> and <i>Leucostethus fraterdanieli</i>. Regarding the reptile population, although results were not entirely conclusive, they were positive. In the latest monitoring, a new species was identified, as well as high secondary type vegetation beneficial to reptile diversity.
Areas needed for habitat connectivity	The EMP of the project establishes the Program 7 “Aquatic ecosystem resource”, which includes the management of ecological corridors to manage the riverine ecological corridor, which is significant because it is the pathway that allows genetic flow through species migration. Also, it is relevant to establish measures for the prevention, mitigation, and compensation of impacts /19//21//23-25/.

Evidence gathering activities, evidence checked, and assessment conclusion	
Habitats for rare, threatened, and endangered species	The Hydroelectric Project El Edén identified risks on altering the habitat used by species as <i>Rheobates palmatus</i>, <i>Bolitoglossa ramosi</i>, and/or <i>Anolis tolimensis</i>, or a possible local extinction of these species. The project includes the following reviewed measures to mitigate the identified risks in the PMS /20-21//23-25/: The Program 3 “Monitoring of endangered and/or endemic species in the direct influence area” includes the monitoring of the populations of the frog <i>Rheobates palmatus</i>, the salamander <i>Bolitoglossa ramosi</i>, and the lizard <i>Anolis tolimensis</i>, which are species endemic to Colombia. The project identifies potential habitats used by these species and their specific requirements, considering vegetation cover and the presence of water bodies. Periodic censuses and monitoring on species are conducted to estimate population sizes and resource use, to apply measures as the following: - Visual censuses every two years during the operation phase. - Population coverage percentages to estimate the population size, and habitat use of the bird species <i>Habia cristata</i> in forest for road construction, among other purposes. - Meetings with the local population twice a year during construction to verify sightings of the species in the region. - Every three months, a local environmental promoter surveys the entire trail system and assess the population status through visual censuses, determining presence or absence of species.
Areas for habitat connectivity	The Hydroelectric Project El Edén identified risks on altering the ecosystem connectivity due to the reduction of flow in the affected section. The project underlines the following measure in the PMS to mitigate the identified risks /20-21//23-25/: Program 4 “Monitoring and follow-up of the aquatic ecosystem” includes the monitoring of conservation of ecological flow, to establishes actions to maintain an ecological flow that preserves the useful habitat and aquatic corridor for the fauna and flora present in the La Miel River between the intake and discharge sections.

4.2.8.2 Introduction of Species

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
N/A	N/A

Existing invasive species	Evidence gathering activities, evidence checked, and assessment conclusion
N/A	N/A

Evidence gathering activities, evidence checked, and assessment conclusion	
Invasive species	N/A

4.2.8.3 Ecosystem conversion

Item	Evidence gathering activities and evidence checked
Ecosystem conversion:	The project covers installation of a hydroelectric plant to provide renewable energy and does not contain any ARR, ALM, WRC or ACoGS components.

4.3 Accuracy of Reduction and Removal Calculations

The verification team has reviewed the ERs spreadsheet provided by the PP /3/ and all the formulae applied, and deems they are in line with the monitoring plan of the registered VCS PD and the applied monitoring methodology /46/.

The monitored parameters used in the calculation of emission reduction of the projects are described below:

Data - Parameter	EG _{facility,y}
Data unit	MWh
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Electricity meter(s)
Description of measurement methods and procedures to be applied	As indicated in the registered VCS PD and the MR /1-2/: the meters installed are bi-directional, taking into account both the quantity of electricity supplied by the project plant to the grid and the quantity of

	electricity delivered to the project plant from the grid. In this case the individual values are not required. Monitoring procedures are implemented onsite or remote, using tele-metering technology. The operational team is in charge of taking the measurements and reporting to XM, as per this monitoring period: 01-January-2021—31-July-2024 /4-5//7/.												
Frequency of monitoring-recording	Continuous and at least monthly recording. Typically, the measured data is read once every 24 hours using tele-metetering technology (remotely) and reported to XM (grid operator and administrator).												
Value(s) monitored	The value monitored for the monitoring period is as follows /4/: <table border="1"> <thead> <tr> <th>Period</th><th>EG_{facility,y} (MWh)</th></tr> </thead> <tbody> <tr> <td>01/01/2021 – 31/12/2021</td><td>86,661.50</td></tr> <tr> <td>01/01/2022 – 31/12/2022</td><td>92,259.73</td></tr> <tr> <td>01/01/2023 – 31/12/2023</td><td>53,153.45</td></tr> <tr> <td>01/01/2024 – 31/07/2024</td><td>33,152.63</td></tr> <tr> <td>Total</td><td>265,227.31</td></tr> </tbody> </table>	Period	EG _{facility,y} (MWh)	01/01/2021 – 31/12/2021	86,661.50	01/01/2022 – 31/12/2022	92,259.73	01/01/2023 – 31/12/2023	53,153.45	01/01/2024 – 31/07/2024	33,152.63	Total	265,227.31
Period	EG _{facility,y} (MWh)												
01/01/2021 – 31/12/2021	86,661.50												
01/01/2022 – 31/12/2022	92,259.73												
01/01/2023 – 31/12/2023	53,153.45												
01/01/2024 – 31/07/2024	33,152.63												
Total	265,227.31												
Monitoring Equipment	During the inspection and through document review, the auditing team has been able to confirm the technical specifications represented below. A main and a backup meter were installed, both bi-directional and with the following characteristics for both meters /8//39-44/: - Brand: Schneider Electric - Model: Power Logic ION8650 socket meter (2011) - Accuracy class: 0.2S - Digital fault recording: Simultaneous capture of voltage and current channels for sub-cycle disturbance transients. - Main meter serial number: MW-1509A351-02 - Backup meter serial number: MW-1509A352-02 Monthly joint meter readings of the main and back-up meters located at project activity sub-station feeders are taken by the designated officials of the company and state electricity utility as confirmed during the on-site visit. The summation of all connected meters reading is used for billing and emission reduction calculation purpose.												

QA-QC procedures to be applied	Calibration dates of the energy meters have been as follow:		
	Description	Main Meter	Back-Up Meter
	Serial Number	MW-1509A351-02	MW-1509A352-02
	Calibration date	15-September-2018	15-September-2018
	Valid till	14-September-2021	14-September-2021
	Calibration date	29-October-2022	29-October-2022
	Valid till	28-October-2025	28-October-2025
	The calibrations were held by laboratory Veritest Ltda. accredited by ONAC under accreditation code 10-LAC-032. According to the registered VCS PD, “The meters will be calibrated every three years based on national standards”: Colombian standard NTC 4,856. Nevertheless, there was a delay in the calibration of the meters between the period 15-September-2021 and 29-October-2022. The electricity generated has been adjusted with meter’s error by the PP in the corresponding ER spreadsheet (tab “El Edén Generation Summary”) in a proper manner /3/. The calibration certificates were checked, and the audit team confirms they are in line with the registered VCS PD and the adjustment is conservative according to the methodology applied and VCS Standard /39-44/. This way, both meters are in compliance with the host country calibration regulations and had valid calibrations during the entire monitoring period.		
Purpose of the data	Calculation of Baseline emissions		
Calculation method	N/A		
Comments	As confirmed during the visit in January 2025, all data collected as part of the monitoring process is archived electronically and kept at least for two years after the end of the last crediting period, according with the registered VCS PD.		

Baseline emissions:

The calculation of the actual net GHG reductions is calculated in line with the ACM0002, version 13.0.0 as follows /3//46/:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

- BE_y = Baseline emissions in year y (tCO₂/y)
- $EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/y)
- $EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh)

For the Quantity of net electricity generation, as the project activity is a green field project, $EG_{PJ,y}$ is the same as $EG_{facility,y}$. This is according to Eq. number 7 of the ACM0002, version 13.0.0.

For the Combined margin CO₂ emission factor ($EF_{grid,CM,y}$), it is calculated ex ante according to the Tool to calculate the emission factor for an electricity system” (version 3.0.0). The value applied for the entire Monitoring Period is 0.3072 tCO₂e/MWh /2-3/.

Project Emissions:

No project emissions are considered for the project activity as the project activity has no reservoir with power density greater than 10 W/m², as requirement of the methodology /46/. Furthermore, no leakage is also considered according to the applied methodology. No deviations have been identified.

Leakage:

As per the methodology ACM0002, version 13.0.0 and as defined in the registered VCS PD, no leakage is considered in the project activity and the same is followed in this monitoring period.

The verification team confirms that:

- According to the applied methodology, the conservativeness of the achieved emission reduction was checked, and the detailed emission reduction calculation has been transparently provided in the ER spreadsheet, which provides sufficient information to allow the reader to reproduce the calculation /3/.
- The information regarding the difference between the ex-ante estimation of ERs presented in the project description and the ex-post calculations presented in the monitoring report, has been provided in section 5.4 of the updated MR. The ex-ante estimation of ERs for the applicable monitoring period are 114,951 tCO₂e, while in the ex-post scenario are 81,476 tCO₂e, being a difference of 33,475 tCO₂e, which represents around 29.12% of lower ERs. The PP has justified the latter fact based on lower electricity generation than expected.

As renewal energy projects depend on natural resources, the audit team considers there are variations not able to be controlled, especially when these estimates are based on initial assessments during the validation stage, from studies that can be slightly different from the reality. As the emission reductions achieved are not higher than the estimations ex-ante, the verification team concludes that justification provided by the PP is plausible and acceptable.

Therefore, the audit team concludes:

- The assumptions, emission factors and default values applied in the calculations are correct and have been properly justified.
- The data has been measured directly from meters and it was cross checked with the supporting documentation provided by the PP /39-44/.
- All the formulae have been found appropriately applied in the GHG emission removals calculations.
- The GHG emission removals and reductions spreadsheets are transparent and clearly referenced.
- The GHG calculation is correct, accurate, traceable and the adjustment applied for the delay in the calibration certificate indicated is conservative and in line with the methodology applied and the VCS requirements /3-5/.

4.4 Quality of Evidence to Determine Reductions and Removals

For the evidence assessment the following activities have been performed:

- Evaluation of all calibration certificates of the main and backup meters /39-44/.
- Assessment and inspection of all installed meters to confirm that they are state of the art and bidirectional.
- Review of the energy generation directly from the SCADA system to cross check the information during the inspection.

Calibration frequency is determined by the monitoring plan stated in the VCS PD of the project, where it is established that the requirement of calibration frequency is three years.

All relevant documents were checked to assess the correctness and quality of data submitted by the project participants, which are used to determine emission reductions.

The records needed for monitoring are archived in line with the requirements of the registered monitoring plan. No significant lack of evidence and missing data were detected during the on-site visit of January 2025. Hence, the verification team confirms that the monitoring system ensures required quality of the monitoring system to ensure the quality of the monitored data. All internal data are subjected to QA/QC measures. The monitoring parameters have been measured/determined without material misstatements and is in line with all applicable standards and relevant requirements /45-53/. The information inflow (from data generation, aggregation, to recording, calculation and reporting) is included above under each parameter and confirms to the requirement of the registered VCS PD. The energy data are measured by the electricity meters and recorded continuously. The data is then reported annually on the VCS Monitoring Report as verified by the verification team through the inspection. Also, it was verified through the audit visit that the operators of the plant have training courses and are sufficiently experienced to carry out the relevant tasks for the operation and maintenance of the hydroelectric plant /54-61/.

Relevant documents such as calibration certificates, electricity generation BBDD, training courses, availability control between the staff members, among others, were verified during the on-site visit. The

verification team finds to be sufficiency of quantity, and appropriateness of quality, of the evidence used to determine the GHG reductions and removals. The audit team considers the quality of the data reported and supporting documentation provided, reliable, solid and adequate level to assure confidence in the accurate quantification of the emission removals and reductions.

4.5 Non-Performance Risk Analysis

Not applicable for the present project activity.

5 VERIFICATION OPINION

5.1 Verification Summary

AENOR has verified that the Hydroelectric Project El Edén is in compliance with the Verified Carbon Standard version 4.7 without qualifications or limitations /45/. The project is located in Colombia, and it is not a grouped project.

The project proponent, ALLCOT AG, has demonstrated that the ownership of the project is unconditional, undisputed, and unencumbered, in accordance with VCS requirements.

The verification of the ex-post emission reductions of the Hydroelectric Project El Edén has been conducted by AENOR in accordance with ISO 14064-3:2019.

AENOR has performed the verification of this project in Colombia on the basis of all issues and criteria of VCS. The conclusions of this report showed that the project, as it was described in the Monitoring Report, is in line with all criteria applicable for the verification.

The verification consisted of the following three phases: i) a desk review of the monitoring report, monitoring plans and supporting documentation; ii) follow-up interviews with project staff and on-site visit; iii) the resolution of outstanding issues and the issuance of the final Verification Report and opinion. In the course of the verification process, corrective actions and clarifications were raised. All have been successfully closed as explained in the verification protocol annexed to this report (Appendix III).

The methodology CDM: ACM0002 - Consolidated baseline methodology for grid-connected electricity generation from renewable sources, Version 13.0 was applied to determine the GHG net anthropogenic reductions of the project /46/.

The review of the monitoring report documentation; the subsequent background investigation, follow-up on-site visit and review of comments by parties have provided AENOR with sufficient evidence to verify the fulfillment of the stated criteria.

In detail, the conclusions can be summarized as follows:

- The project is in line with all relevant criteria of VCS Standard v4.7 and VCS Program Guide v4.4 /45//51/.
- The project is in line with the VCS Templates: v4.4 for Monitoring Report and v4.4 for the Verification Report /49-50/.
- The analysis of the baseline emission, project emissions and leakage has been carried out in a transparent and conservative manner /3-5/.
- The total emission reduction calculated by the project proponent for this monitoring period is 81,476 tCO₂e. The audit team has assessed all the calculations and deems them as correct.

5.2 Verification Conclusion

As previously commented in section 5.1, the total emission reduction of this monitoring period, is 81,476 tCO₂e.

The project involves the ERs through renewable hydraulic energy in the host country Colombia, and the emissions have been calculated in line with the CDM consolidated methodology ACM0002 - Consolidated baseline methodology for grid-connected electricity generation from renewable sources, version 13.0.0 applied by the project.

The VCS Verification was based on the draft monitoring report, emission reduction calculation spread sheet, the monitoring plan as set out in the registered VCS PD and supporting documents made available to the AENOR CONFIA S.A.U. by the project participant.

Verification period: From 01-January-2021 to 31-July-2024

Verified GHG emission reductions and carbon dioxide removals in the above verification period:

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCU (tCO ₂ e)	Removal VCU (tCO ₂ e)	Total VCUs (tCO ₂ e)
1 Jan 2021 - 31 Dec 2021	26,622	0	-	26,622	-	26,622
1 Jan 2022 - 31 Dec 2022	28,342	0	-	28,342	-	28,342
1 Jan 2023 - 31 Dec 2023	16,328	0	-	16,328	-	16,328
1 Jan 2024 - 31 Jul 2024	10,184	0	-	10,184	-	10,184
Total	81,476	0	-	81,476	-	81,476

AENOR considers that the project will likely achieve the calculated GHG emission reductions as they have been stated alongside the Monitoring Report (MR) and this Verification Report /1//3/.

It is opinion of AENOR, with a reasonable level of certainty, that the reasonableness of assumptions, the described limitations, the methods that supported each one of the claims about the outcome of activities and the claimed calculated ex-post emission reductions are free from major errors, omissions, or inaccuracies.

5.3 Ex-ante vs Ex-post ERR Comparison

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
1 Jan 2021 - 31 Dec 2021	32,102	26,622	-17.07%	Lower-than-expected electricity generation due to lower-than-anticipated hydrological resources and the event mentioned in Section 3, which took the plant out of operation for 10 days.
1 Jan 2022 - 31 Dec 2022	32,102	28,342	-11.71%	Lower-than-expected electricity generation due to lower-than-anticipated hydrological resources.
1 Jan 2023 - 31 Dec 2023	32,102	16,328	-49.14%	Lower-than-expected electricity generation due to the El Niño phenomenon, which significantly reduced water availability.
1 Jan 2024 - 31 Jul 2024	18,645(*)	10,184	-45.38%	Lower-than-expected electricity generation due to the El Niño phenomenon, which significantly reduced water availability.
Total	114,951	81,476	-29.12%	

(*) ERs of corresponding period calculated as follows:

Amount of days between 01-Jan-2024 to 31-Jul-2024: 213- 58% of calendar year.

Ex-ante estimated reductions: 32,102 tCO_{2e} * 58%= 18,645 tCO_{2e}

Madrid, 20th November 2025.



Asis Arranz Arbex
Lead Auditor

APPENDIX I: COMMERCIALLY SENSITIVE INFORMATION

Section	Information	Justification	Assessment conclusion	method	and
N/A	N/A	N/A	N/A		

APPENDIX II: LIST OF EVIDENCE PROVIDED

N°	Documents reviewed or referenced
	MR
1	VCS MR Project ID 1068 01012021 - 31072024 V6
	PD
2	VCS Project Description Hydroelectric Project El Eden (16Sept13)
	Calculations
3	El Edén ER MP2_V3
4	Electricity generation 2021 - 2024
5	XM first generation support
	Ownership
6	Merger resolution
7	Eden OC Declaration - XM
8	Andritz contracts signed
9	Andritz end commissioning certificate
10	Connection contracts
	Stakeholder consultation
11	Visit report – Eisaura Lopez
12	Environmental Compliance Reports
13	Environmental Compliance Reports settled and signed
14	Requests, Complaints, Claims and Suggestions (RCCS) records
15	5. Inf-023-Characterization (section 3.4.1)
	Environment
16	Environmental Impact Assessment
17	Environmental License
18	Environmental Impact Assessment resolutions
19	Environmental Management Plan – Operation Phase
20	Environmental management plan form - biotic
21	Environmental management plan form - physical
22	Environmental management plan form - social
	Wildlife monitoring
23	Fauna report
24	Monitoring report - PPT
25	Water, air, fauna, noise and socioeconomic reports
	Internal policies
26	ANHQ-03 QUESTIONNAIRE FOR THE CHARACTERIZATION OF ETHNIC COMMUNITIES
27	ANHQ-05 Human Rights Due Diligence Protocol
28	ANHQV-06 GUIDELINES FOR MANAGING THE CONTRACTORS' ENVIRONMENT
29	GTH A 04 WORK DISCONNECTION - PRO
30	GUHQ-03 GUIDE TO RELATIONSHIPS WITH ETHNIC COMMUNITIES

N°	Documents reviewed or referenced
31	PDHQ-08 UPDATE ACTION IN CASE OF BLOCKAGE OR VIOLATION OF DEFECT
32	PDHQ-065 RECEIPT AND RESPONSE TO PETITIONS, COMPLAINTS, CLAIMS, CONGRATULATIONS AND SUGGESTIONS - PQRFS
33	PDHQ-066 FILING COMPLAINTS AND/OR CLAIMS
34	PODE-03 CORPORATE SOCIAL RESPONSIBILITY POLICY
35	PODE-04 NON-DISCRIMINATION AND CHILD LABOR
36	PODE-05 HUMAN RIGHTS POLICY
37	PODE-09 CODE OF ETHICS
38	RETH-01 INTERNAL WORK REGULATIONS PRO
	Calibration certificates
39	ME-1809-22785
40	ME-1809-22786
41	ME-1809-22787
42	ME-1809-22788
43	ME-2210-30483
44	ME-2210-30484
	Methodologies and tools
45	VCS Standard, v4.7
46	CDM ACM0002, Consolidated baseline methodology for grid-connected electricity generation from renewable sources, v13.0.0
47	CDM Tool01, Tool for the demonstration and assessment of additionality, v7.0
48	CDM Tool07, Tool to calculate the emission factor for an electricity system, v3.0.0
49	VCS-Monitoring-Report-Template-v4.4
50	VCS-Verification-Report-Template-v4.4
51	VCS-Program-Guide-v4.4
52	VCS-Program-Definitions-v4.5
53	VCS-Methodology-Requirements-v4.4
	Other supporting documentation
54	O&M daily reports 2021 - 2024
55	Pedro Arcilla contract
56	18.1. CHEE_Torrential Flood Report_V0 May 2021
57	Environmental management plan - PPT
58	Training areas of influence
59	Political socialization
60	Training records
61	Assets and withdrawals of SDG 8 2021 - 2024
62	On-site interviews El Edén
63	AENOR questionnaire
64	Communications-Agreement
65	VCS-Accession-Representation
66	VCS-Partial-Release-Representation-1068

APPENDIX III: FINDINGS

Corrective Actions Requests (CARs)

NC/CAR id.	01	Date: 24/01/2025
NC/CAR description		
The audit team has identified the following issues in the Monitoring Report (MR) that are not consistent with the VCS MR Template version 4.4: 1. Title of the project is not reflected in the cover page. 2. The “Original data of issue” is pending to be filled in the cover page. 3. The VCS Standard Version reflected in the cover page is not correct. 4. The index is not consistent with the template. 5. The total GHG emission reductions and carbon dioxide removals generated in the monitoring period are not represented in section 1.1. 6. Template tables displayed for sections 2.1.3, 2.3.5 and 2.4.2 are missing. 7. After section 2.4.3, the number of several sections are not correct, in line with the template form. 8. In sections 3.1 and 3.2, in the table row of “Justification of choice of data or description of measurement methods and procedures applied”, some detailed information related to the responsible person/entity that undertook the measurement, and the date of the measurement, is missing. 9. More detailed information on relevant equations is missing on section 4.4. 10. APPENDIX 1: COMMERCIAL SENSITIVE INFORMATION is missing.		
Project proponent’s response		Date: 25/02/2025
Items 1, 2, 3, 4, 6, 7, 8, 9, and 10 were solved. Regarding item number 5, the last tense of the last paragraph of section 1.1. states “reducing a total of 81,476 tCO₂ during the same period. “		
Documentation provided by the project proponent		
VCS MR Project ID 1068 01012021 – 31072024 V2		
VVB’s evaluation		Date: 05/03/2025
Item 4 is still incorrect, format and titles. In addition, rows of “Most recent date of issue” and “Version” in the cover page have not been updated appropriately. Therefore, CAR 01 is still open.		

Project proponent's response	Date: 06/03/2025
Index was corrected. Most recent date of issue" and "Version" were updated.	
Documentation provided by the project proponent	
VCS MR Project ID 1068 01012021 – 31072024 V3	
VWB's evaluation	Date: 13/03/2025
Issues indicated have been addressed by the PP correctly. Therefore, CAR 01 is closed.	

NC/CAR id.	02	Date: 24/01/2025
NC/CAR description		
The audit team has identified the following errors in the MR: 1. The website provided in page 28 “www.xm.com.co” does not work. 2. “Percent difference” represented in section 4.4 are not correct.		
Project Proponent’s Response		Date: 25/02/2025
The website is working. Please try again. Percents were corrected.		
Documentation provided by the project proponent		
VCS MR Project ID 1068 01012021 – 31072024 V2		
VWB’s evaluation		Date: 05/03/2025
1. The website still has problems, when using the 'Ctrl+left mouse button' keys you should be able to access it directly: 403 Forbidden Transaction ID: 4a05ddaa5210fdd4f351412435042c19b9f3d580c0b6ee133588c9a6aa86b0ca		

2. Percent difference of the second table in Section 5.4 of the MR, have been updated and now are correct. In addition, achieved reductions have been correctly recalculated for the 2021 and 2022.

CAR 02 is still open.

Project Proponent's Response

Date: 06/03/2025

In my case, when using the "Ctrl+left button" keys, link directs me to the website.



Documentation provided by the project proponent

VCS MR Project ID 1068 01012021 – 31072024 V3

VVB's evaluation

Date: 13/03/2025

After internal searching, it seems there are countries where the website represented does not work. As the website shall be public and with access to all users, the VVB suggest reflecting in the MR the following link: <https://sinergox.xm.com.co/Paginas/Home.aspx>

CAR 02 is still pending to be resolved.

Project Proponent's Response

Date: 13/03/2025

The link has been changed.

Documentation provided by the project proponent

VCS MR Project ID 1068 01012021 – 31072024 V4

VVB's evaluation

Date: 19/03/2025

Requested public website has been modified correctly in the updated MR v4.

Therefore, CAR 02 can be closed.

NC/CAR id.	03	Date: 24/01/2025												
NC/CAR description														
The capacities of the installed turbines and generators specified in the registered PD (version 04, dated on September 16th/2013) differ from those confirmed during the onsite visit: <table> <tr> <th></th><th>Turbines MW</th><th>Generators MWA</th></tr> <tr> <td>Registered PD</td><td>10.115</td><td>11.174</td></tr> <tr> <td>MR, version 01</td><td>10.319</td><td>11.154</td></tr> <tr> <td>OnSite inspection</td><td>10.319</td><td>11.154</td></tr> </table>				Turbines MW	Generators MWA	Registered PD	10.115	11.174	MR, version 01	10.319	11.154	OnSite inspection	10.319	11.154
	Turbines MW	Generators MWA												
Registered PD	10.115	11.174												
MR, version 01	10.319	11.154												
OnSite inspection	10.319	11.154												
Similarly, according to the installed capacity of the generators and their power factor, the installed capacity of the hydroelectric plant is 20.078 MW, which differs from the installed capacity declared in the PD and the MR.														
Project Proponent's Response		Date: 25/02/2025												
<i>The PD states that "The installed capacity of the plant at the interconnection point is 19.5 MW (this is based on the design and can vary slightly under real conditions)." In this context, the identified difference between the turbine and generator capacity has already been addressed in the Deviations (3.2.2) section.</i> <i>These modifications are not considered significant, as they do not affect either the baseline or additionality. In the latter case, a sensitivity analysis was included in the registered PD, considering a scenario of up to 30 MW.</i> <i>The justification is incorporated into the MR.</i>														
Documentation provided by the project proponent														
VCS MR Project ID 1068 01012021 – 31072024 V2														
VVB's evaluation		Date: 05/05/2025												

The issue has been explained by the Project Participant (PP) and the MR has been updated accordingly. Nevertheless, no information has been added according to the installed capacity of the hydroelectric plant by 20.078 MW. In addition, no accurate reference has been added on the sensitivity analysis included, considering a scenario up to 30 MW. Therefore, CAR 03 is still pending to be resolved.	
Project Proponent's Response	Date: 06/03/2025
An error was made. The sensitivity analysis includes a scenario with a 10% increase in generation, raising the projected output from 104.5 GWh to 114.95 GWh, which would require increasing the capacity from 20.11 MW to 22.12 MW. The 30 MW capacity was considered solely for the common practice analysis. These considerations were added to MR.	
Documentation provided by the project proponent	
VCS MR Project ID 1068 01012021 – 31072024 V3	
VVB's evaluation	Date: 13/03/2025
PP has added the information described relating the sensitivity analysis on Section 3.2.2 Project Description Deviations of the MR, in line with sub-step 2d of Section 2.5 of the registered PD. In addition, this generation increased complies with the “Guidelines on Common Practice” (version 02.0) provided in Annex 8 of the EB Report 62, which has been applied for the assessment of common practice analysis of the project. The step 1 of the latter indicates the following: “Step 1: Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity”. CAR 03 can be closed.	

Clarification requests (CLs)

CL ID	01	Date: 24/01/2025
CL description		
The audit team has determined that the following evidence and supporting documentation must be provided by the project participant (PP): 1. How the project started to vert electricity to the National Interconnected System (SIN) since 16/02/2017. 2. Certificate of Compliance with Connection Code, where the installed capacity of the plant at the interconnection is displayed. 3. Signed contract for the supply, assembly and commissioning of electromechanical equipment. 4. End of Commissioning Certificate. 5. Evidence of the first round of stakeholder consultation in August 2010. 6. Procedure that has the project for the receipt and response of requests, complaints, claims, compliments, and suggestions (PQRFS). 7. Compliance Reports where stakeholder inputs can be contrasted. 8. Environmental compliance reports (ICAs by its acronym in Spanish) years: 2021, 2022, 2023. 9. Evidence of delivering those Environmental compliance reports previously mentioned. 10. Wildlife monitoring performed during 2022. 11. Environmental Impact Study. 12. Environmental management plan. 13. Environmental license granted by CORPOCALDAS, and its modifications. 14. Ownership of the project. 15. Contract of a non-qualified worker used for the construction is locally hired. 16. Training records. 17. Internal policies: for work regulations, non-discrimination and child labor prevention and human rights. 18. Stops related to the electrical equipment described in section 2.5. 19. Calibration certificates for MW-1509A351-02 and MW-1509A352-02 meters, during 2018 and 2022.		
Project proponent's response		Date: 25/02/2025
1. <i>The End of Commissioning Certificate was signed on the 28/February/2017. However, according to the SIN administrator, XM, the project Hydroelectric El Edén has been verting electricity to the SIN, on the 16-February-2017. Injection data obtained from the administrator's website, prior to the commissioning date, is considered evidence.</i> 2. <i>According to AGREEMENT 646 of the NATIONAL OPERATION COUNCIL (CNO), the plant is the one that declares the effective power used for commercial purposes. In this sense, El Edén has declared a power of 19.9 MW as indicated in the attached note and in the capture of the national administrator's information system. It is worth mentioning that the power of 19.9 MW is the maximum that is delivered on the commercial border, in the substation located in Marquetalia. A support file is attached.</i> 3. <i>A support file is attached.</i>		

4. A support file is attached. See section 3.4.1. of EIA
5. A support file is attached.
6. A folder is attached containing the Environmental Compliance Reports (ICAs) and the consolidated record of requests, inquiries, and complaints.
7. ICAs are included in the folder 7.
8. A folder is attached containing the ICA filings.
9. The information related to the wildlife monitoring conducted in late 2021 is attached. It is worth mentioning that no monitoring was carried out in 2022. The next monitoring was completed in early 2024. The corresponding report is attached.
10. A folder containing information related to the EIA is attached.
11. A folder containing information related Environmental management plan is attached.
12. Information is contained in folder 11.
13. A support file is attached.
14. A support file is attached.
15. A folder with training records is attached.
16. A folder is attached containing information on internal policies for work regulations, non-discrimination, child labor prevention, and human rights.
17. The most significant adverse event is included in section 2.5. A report with details of the event and daily O&M files are shared.
18. A folder with calibration certificates is attached.

Documentation provided by the project proponent

1. XM first generation support
- 2.1.20170301 DECLARACION OC EDEN (1) 20130625 Contrato Andritz Firmado
- 2.2.Declaración OC Edén
- 3.2.20130710 CH EL EDEN - OTRO SI 1 CONTRATO ANDRITZ
4. puesta en marcha andritz
5. Inf-023-Characterizacion (section 3.4.1)
6. PDHQ-065 RECEPCIÓN Y RESPUESTA DE PETICIONES, QUEJAS, RECLAMOS, FELICITACIONES Y SUGERENCIAS – PQRFS
7. Folder: 7. ICA-Stakeholder report
8. Folder: 7. ICA-Stakeholder repor
9. Folder: 9. Radicados ICA
10. Folder: 10. Wildlife monitoring
11. Folder: 11. Environmetal Impact Assessment
12. Folder: 12. Environmental Management Plan
13. Folder: 11. Environmetal Impact Assessment
14. Resolución Fusion
15. Contract PEDRO ARCILA
16. Folder: 16. Training records
17. Folder: 17. Internal policies
- 18.1. CHEE_Informe avenida torrencial_VO Mayo 2021
- 18.2. INFORMES DIARIOS O&M
19. Folder: 19. Calibration certificates

VVB's evaluation	Date: 05/03/2025
All evidence asked have been provided by the PP, except for the following items: - (2) The clarification provided by the PP is in a different language than English. Besides, the VVB has not been able to find the content information represented. Taking into account that the agreement 646 is from 2013 and the measurement code is from 2014, it seems there is disjointed information. Therefore, clarification and adding reference of the corresponding source of data is required. - (5) Environmental impact assessment evidence provided instead of a stakeholder consultation. - (7 and 8) Seven ICAs for El Edén have been provided. However, it seems the responsible signature gap of the reports are empty and the VVB requires clarification. - (19) Although the reception date of the two calibration certificates provided for the two meters indicated in 2022 is 28/10/2022, both certificates show the calibration certificate on 29/10/2022. Please clarify:  As some issues are still pending to be close, CL 01 is still open.	
Project Proponent's Response	Date: 06/03/2025
(2) Clarification was translated to English. In 2013, National Council of Operation (Consejo Nacional de Operación-CNO) approved Agreement 646, which amended the "Guideline Procedure for the Commissioning of Power Plants into the SIN, Assets of the National Transmission System (STN), the Regional Transmission System (STR), and Connection Assets to the STN" (see link in the section "Documentation provided by the project proponent"). In 2014, the power plant signed Connection Contract No. 046 of 2014, which established the connection conditions for El Edén to the substation. The contract specifies that the maximum capacity allocated to El Edén at the connection point is 20 MW (the folder containing the contract and its amendments is attached). In February 2017, the project began injecting electricity (evidence has already been provided through the generation data obtained from XM). On March 2, 2017, it self-declared commercial operation, with a maximum capacity at the connection point of 19.9 MW, in accordance with Agreement 646 of the CNO (a screenshot indicating this, the note sent to XM's administrator, and a link to the published data on the administrator's website are attached). 5) The file "Inf-023-Characterizacion " is part of the EIA and the Stakeholder Consultation evidence is provided in its section 3.4.1. It is important to highlight that the public consultation was conducted by the construction company, not for the current PP. Considering the time that has elapsed, obtaining	

additional supporting documentation is likely to be challenging.

3.4.1.1 Municipio de Pensilvania

En este municipio se realizaron dos reuniones diferentes: una de ellas tuvo lugar el 29 de octubre de 2010 en el despacho de la alcaldía del municipio de Pensilvania, a esta asistieron algunos representantes de la administración como son: el Coordinador del Área de Desarrollo, el Jefe de Control Interno, el Inspector de Policía y el Ingeniero encargado de la parte ambiental del municipio, para un total de 4 personas. Con ellos se trataron



Fotografía 3.43 Reunión con representantes de la administración municipal de Pensilvania



Fotografía 3.44 Participantes a la reunión del corregimiento Bolivia



Fotografía 3.46 Reunión alcaldía de Marquetalia

(7) and (8) The ICAs are not signed within the document itself but are instead submitted alongside an official letter signed by the Central's legal representative. Supporting letters are attached as evidence.

(19) Dates were modified to 29/10/2022. Besides, ER spreadsheet was updated adding the adjustment to the record of 29/10/2022.

Documentation provided by the project proponent

VCS MR Project ID 1068 01012021 – 31072024 V3

El Edén ER MP2_V3

(2)

Link to Agreement 646 of the CNO:
https://gestornormativo.creg.gov.co/gestor/entorno/docs/acuerdo_cno_0646_2013.htm

Folder: CONTRATO DE CONEXION

Folder: Commercial operation declaration and link to effective power published on the XM website

Follow the following link and route: <https://paratec.xm.com.co/reportes/capacidad-efectiva-neta-tipo-generacion>

Planta	Hidráulica/	No	Despachada	Centralmente/	Generador/
ELEDEN		19,9			
EL LIMONAR		1R			

(7) and (8)

Folder: Signed ICAs

VVB's evaluation	Date: 13/03/2025
(2) The evidence has been clarified and referenced correctly by the PP. (5) The PP has clarified the corresponding section of the EIA provided, where information on stakeholder consultations is described. Nonetheless, dates of the consultations reflected seems not to be consistent with the MR. Please clarify. (7) and (8) signed supporting letters representing ICAs have been provided and the VVB deems them correct. (19) Indicated dates have been corrected by the PP adequately, as well as the corresponding adjustment in the ER spreadsheet. As there is still an issue to be solved, CL 01 is still open.	
Project Proponent's Response	Date: 13/03/2025
As indicated in the registry PD, the local stakeholder consultation process began in August 2010. The first round of consultations extended until May 2011. This process involved internal preparation, outreach events, and time for receiving feedback. The outreach events for this first round took place on October 29 and 30, 2010, in Pensilvania; on October 30, 2010, in Manzanares; and on November 4, 2010, in Marquetalia. This clarification is included in the MR.	
Documentation provided by the project proponent	
VCS MR Project ID 1068 01012021 – 31072024 V4	
VVB's evaluation	Date: 19/03/2025
The pending issue has been clarified by the PP. The VVB has reviewed the PD of the project, as well as the information added in the updated MR v4, and deems it acceptable. CL 01 can be closed.	

CL ID	02	Date: 24/01/2025
CL description		
The following issues from the MR require clarification or more detailed information by the Project Participant (PP) to the auditing team: 1. Emails are missing in the tables displayed of section 1.4.		

2. Mentions of the applicable international human rights law and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples, are missing, according to the VCS MR template form. 3. In section 2.4: a. If it is detected that the indicated flora is affected, what type of measures would be used to avoid this negative effect? b. If it is detected that there is soil degradation, what type of measures would be used to avoid this negative effect? c. If it is detected that there is water stress and degradation, what type of measures would be used to avoid this negative effect? d. About species or habitats, how have been the results of the periodic monitoring done during the construction and operation? If results have been negative, what the project does to mitigate negative effects? e. What actions would the project take to improve/balance the indicated species censuses? 4. Dates details on incidents described are missing in section 2.5. 5. Some references as to "Resolution CREG 025 of 1995" (p.24), CREG-038 Resolution (p.27) and Colombian standard NTC 4,856 (p.27), are missing. 6. According to the ERs spreadsheet ("El Edén ER MP2"): "The electricity generated during the period between 15-September-2022 and 28-October-2022 was adjusted with meter's error." However, the VVB has not found these adjustments. 7. According to the PD: "Calibration tasks follow national standards and are in accordance with the calibration instructive specified in Colombian standard NTC 4,856 for electricity metering devices. There is no regulatory requirement for calibration frequency. The meters will be calibrated maximum every three years based on national standards". Nevertheless, the MR indicates in section 3.2 that "The frequency of calibration is four years according to CREG-038 Resolution". 8. Some equation references with their corresponding methodology are missing in section 4.1, as well as some parameter's description.	
Project proponent's response	Date: 25/02/2025
1. <i>Emails were included.</i> 2. <i>Required mentions were included.</i> 3. <i>Measures were detailed in the document. Regarding item e), monitoring is conducted solely for tracking purposes. If a decline in any species is detected, the environmental authority is responsible for determining the appropriate course of action. El Edén does not take any direct measures, as all decisions and actions regarding wildlife management fall under the exclusive jurisdiction of the environmental authority, given the complexity of its regulations in this field.</i> 4. <i>Data is included.</i> 5. <i>References were added to MR. CREG-038 Resolution is deleted since the calibration frequency is established as per Monitoring Plan.</i> 6. <i>A new version of the ERs spreadsheet is shared. Please, refers to green highlighted cells in the sheet "El Edén Generation Summary"</i> 7. <i>The regulation Resolution CREG-038 dates back to 2014, after the development of the PD, and states that the calibration requirement is every four years. It is acknowledged that the</i>	

<i>PD monitoring plan establishes calibration every three years, so this frequency will be maintained, as it is more conservative than the current regulation. To be conservative in estimating emissions, the meter error will be deducted from the records between September 14, 2021, and October 28, 2022.</i> 8. <i>Equation references and parameter's description are included.</i>	
Documentation provided by the project proponent	
VCS MR Project ID 1068 01012021 – 31072024 V2 EI Edén ER MP2_V2	
VVB's evaluation	Date: 05/03/2025
1. Missing gaps filled correctly in the MR updated. 2. Information required added appropriately by the PP in the MR (Section 2.3.2). 3. Section 2.4 has been updated with much more detailed information regarding measures applied by the project for the ecosystem health. The auditing team has assessed the content information added in the MR, as well as all the supporting documentation provided by the PP, and deems it complete and true. Nonetheless, some of this information has been added in other language than English and in a different format with white colour letter. Therefore, this section requires to be revised by the PP. 4. Section 2.5 of the MR has been updated with the required content information. 5. Indicated references added appropriately in the MR. 6. Adjustments verified by the audit team, who deems them conservative, consistent and in line with the monitoring plan. 7. The most conservative current regulation has been applied by the PP in line with the monitoring plan by calibration frequency of three years. The meter error adjustments for the estimating emissions have been verified by the VVB who deems them acceptable. However, as commented in CL 01-19 before, it is pending to be clarified by the PP if the calibration date was in 28/10/2022 or in 29/10/2022. 8. Equation references have been added in the MR adequately. However, reference equation for “Power density of the project activity (W/m²)” is still missing and could be reflected. Pending to close few items, CL 02 is open.	
Project Proponent's Response	Date: 06/03/2025
(3) Information in Spanish has been translated to English and the format was revised. (7) The calibration was in 29/10/2022. Dates were modified in the MR. (8) Reference equation was added.	
Documentation provided by the project proponent	
VCS MR Project ID 1068 01012021 – 31072024 V3	

VVB's evaluation	Date: 13/03/2025
(3) The indicated information of Section 2.4 of the MR has been modified in the correct language. However, when track changes is removed, the VVB has still identified some typos and white letter in Section 2.4.1. (7) As commented in the previous finding CL 01-19, this issue has been solved. (8) Indicated methodology equation has been referenced adequately by the PP in the MR updated. CL 02 is still open.	
Project Proponent's Response	Date: 13/03/2025
The section was revised.	
Documentation provided by the project proponent	
VCS MR Project ID 1068 01012021 – 31072024 V4	
VVB's evaluation	Date: 19/03/2025
Section 2.4 has been fully revised by the VVB and deems it correct. Therefore CL 02 can be closed.	

Forward action request (FAR)

FAR ID	01	Date: 20/11/2025
FAR description		
The PP is requested that, as an integral part of the verification plan for the future verification periods, a specific assessment designed to confirm the absence of double counting or double claiming associated with Colombia VCS projects.		
Project proponent's response		Date: DD/MM/YYYY
Documentation provided by the project proponent		
VVB's evaluation		Date: DD/MM/YYYY